

THOMAS PULICKAL JOSEPH

SENIOR SOFTWARE ENGINEER

WEST CHESTER, PA

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Professional Summary

Senior Software Engineer with experience building high-throughput automation platforms, real-time control systems, and distributed software solutions for consumer electronics manufacturing. Specialized in C#, ASP.NET Core, Beckhoff TwinCAT, robotics integration, and scalable backend systems. Proven track record designing production-grade automation software, concurrent communication frameworks, and telemetry platforms that improve throughput, reliability, and operational efficiency.

Experience

COMMUNICATIONS TEST DESIGN, INC

SENIOR SOFTWARE ENGINEER

JAN 2026 – PRESENT

West Chester, PA

- Developed a real-time C# communication framework using the Task Parallel Library (TPL) to bridge enterprise software and industrial hardware, sustaining over 1,000 transactions per second through TwinCAT ADS/AMS integration.
- Leveraged AI agents (Claude) to architect complex C# and PHP modules, accelerating initial prototyping by 50% while maintaining strict adherence to enterprise design patterns and security standards.
- Optimized the software development lifecycle (SDLC) by integrating AI agents for real-time code auditing and automated test generation, reducing peer-review cycles and increasing overall sprint velocity.
- Designed and developed internal web applications hosted on internal company infrastructure using HTML, PHP, and SQL to support operations.
- Built dynamic data-driven web pages with PHP backend and SQL database integration for real-time production data access.
- Created internal tooling and dashboards for test result visualization and reporting via web interfaces.
- Architected and deployed high-throughput automated test cells for consumer electronics, specifically integrating Beckhoff xPlanar motion platforms with custom fixtures to eliminate manual handling.
- Engineered complex motion sequences using TwinCAT 3 (Structured Text), optimizing mover routing and precision positioning for synchronized electrical signaling and power-on testing.
- Architected and deployed cross-platform telemetry and control systems using Avalonia UI and ASP.NET Core, enabling real-time diagnostics and automated grading workflows.
- Analyzed robotic motion profiles and sensor data to refine cycle times, resulting in a 30% increase in process throughput and a 40% reduction in system downtime.
- Designed and implemented RESTful APIs to synchronize robotic test results with enterprise databases.
- Collaborated cross-functionally with electrical and mechanical engineering teams to translate hardware specifications into scalable software solutions, utilizing Git workflows and Octopus Deploy.
- Authored comprehensive technical documentation, including system diagrams and test procedures, to facilitate seamless maintenance and continuous improvement cycles.

SOFTWARE ENGINEER

AUG 2023 – DEC 2025

West Chester, PA

- Developed modular automation platforms for device handling and semi-robotic operations using C#, ASP.NET Core, and Knockout.js to support CTDI test and repair workflows.
- Developed and maintained web-based phone testing software used across CTDI operations, integrating automation platforms, test equipment, and backend services to support high-volume device processing.
- Designed and maintained web-based control dashboards for real-time visibility into automation status, queues, and device-level test results.
- Implemented REST APIs to integrate automation modules, test sequencers, and internal/third-party systems, enabling reliable data exchange and orchestration of test flows.
- Supported server-side automation using Apache HTTP Server with PHP and SQL for processing test data, generating reports, and monitoring system health.
- Improved stability, efficiency, and maintainability by refactoring legacy services, adding automated validation, and enhancing observability.
- Collaborated in Agile teams using Git and Jira for source control, code review, issue tracking, and controlled deployment of releases into CTDI production environments.

SOFTWARE ENGINEER

Guntur, India

- Developed microservices and device-control APIs for industrial automation systems using C# .NET and PostgreSQL.
- Integrated backend logic for machine-assisted monitoring, improving operational precision and speed.
- Designed optimized SQL schemas and procedures for high-performance analytics.
- Contributed to full-stack projects in Agile Scrum environments for timely delivery of automation software.

ENGINEER'S HUB

JUNE 2019 – AUG 2019

SOFTWARE ENGINEER INTERN

Hyderabad, India

- Built a hospital management system featuring automated patient record handling in PHP and MongoDB.
- Streamlined data exchange workflows and improved user-facing interface reliability.

Technical Skills

- **Programming Languages:** C#, C++, Python, Structured Text (PLC), Java, C, PHP, SQL.
- **Robotics & Industrial Automation:** Beckhoff TwinCAT 3, EtherCAT, xPlanar Motion Control, Real-Time Distributed Systems.
- **Electrical & Electronics:** Digital/analog I/O, sensor and actuator integration, custom test fixture integration and Electrical Test Sequencing.
- **Computer Vision:** Vision-driven inspection for functional grading.
- **Backend & Web Technologies:** ASP.NET Core, Avalonia UI (Cross-platform XAML), JavaScript, Knockout.js, ReactJS, PHP, Apache HTTP Server, RESTful API Development.
- **DevOps & Tools:** Git, GitHub, Octopus Deploy, Postman, Jira, Visual Studio, TwinCAT Scope View.
- **Databases:** PostgreSQL, MySQL, SQLite, MongoDB.
- **Methodologies:** Agile, Scrum, CI/CD, SDLC, TDD (Test Driven Development)

Education

UNIVERSITY OF HOUSTON-CLEAR LAKE

Master of Science in Computer Science (GPA: 3.70 / 4.00)

May 2023

Houston, TX

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY

Bachelor of Technology in Computer Science & Engineering (GPA: 7.75 / 10.00)

Sep 2020

Hyderabad, India

Certifications

Beckhoff TwinCAT 3 Robotics Training (TR3025)

Basic I/O, Communication, System Manager and PLC Programming (Beckhoff Automation, Feb 2025)

IBM Data Science Professional Specialization

Python, data analytics, visualization, and machine learning fundamentals (IBM / Coursera, Jul 2022)

Joy of Computing using Python

Python programming, algorithms, and computational thinking (SWAYAM MHRD / NPTEL, Dec 2019)